Mach-Zehnder Interferometer: Design, Simulation, and Data Analysis

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I. Introduction

A Mach–Zehnder Interferometer (MZI) is a fundamental component widely used in photonic integrated circuits (PICs). It operates by splitting incoming light into two separate optical paths, which introduces a relative phase difference between them, and then recombining the signals to produce interference. In a typical PIC implementation, the MZI consists of two optical splitters (often Y-branches, directional couplers, or multimode interferometers) connected by two waveguide arms. By engineering the optical path length difference between these arms (ΔL), a wavelength-dependent periodic interference pattern is generated. The spacing between adjacent transmission peaks, known as the free spectral range (FSR), is directly related to this path length difference. Additionally, the measured FSR provides a convenient way to extract the group index of the waveguide.

II. Modeling and simulation

A. Waveguide

The waveguide used in this study is a silicon core strip waveguide with a width of $500\,\mathrm{nm}$ and a height of $220\,\mathrm{nm}$, surrounded by SiO_2 cladding (Figure 1).

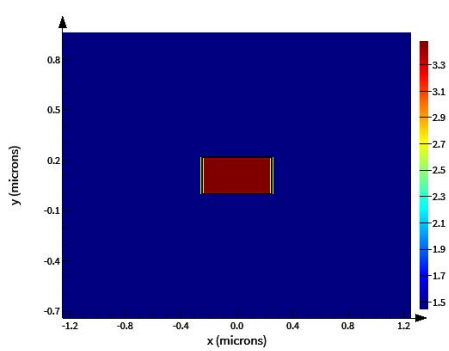

Fig. 1. The cross-section of the waveguide.

Lumerical MODE solver is used to obtain the E-field intensity profile for the fundamental TE and TM mode, as shown in Figure 2 and Figure 3, respectively.

A wavelength sweep from $1500\,\mathrm{nm}$ to $1600\,\mathrm{nm}$ is performed to determine the relationships between the effective index and wavelength, as well as between the group index and wavelength. For TE mode, the effective index decreases as the wavelength increases (Figure 4), while the group index increases as the wavelength increases (Figure 5). For TM mode, the effective index decreases as the wavelength increases (Figure 6), and the group index also decreases as the wavelength increases (Figure 7).

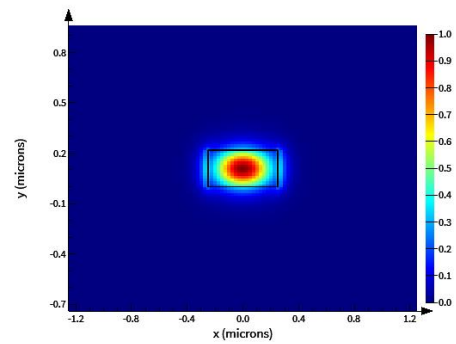

Fig. 2. The simulated waveguide mode profile for TE polarization.

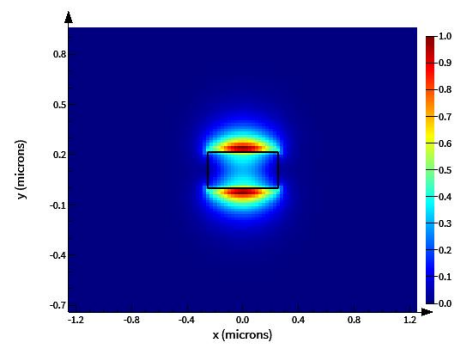

Fig. 3. The simulated waveguide mode profile for TM polarization.

A compact waveguide model can be obtained by fitting the simulated data to a second-order polynomial, which enables prediction of the effective index at a given wavelength. The model is constructed using a Taylor expansion around the central wavelength:

$$n_{\text{eff}}(\lambda) = n_1 + n_2\,(\lambda - \lambda_0) + n_3\,(\lambda - \lambda_0)^2 \tag{1}$$

Using the simulated data from Figure 4 and Figure 6, the Taylor expansion parameters can be extracted for TE and TM modes (Table II).

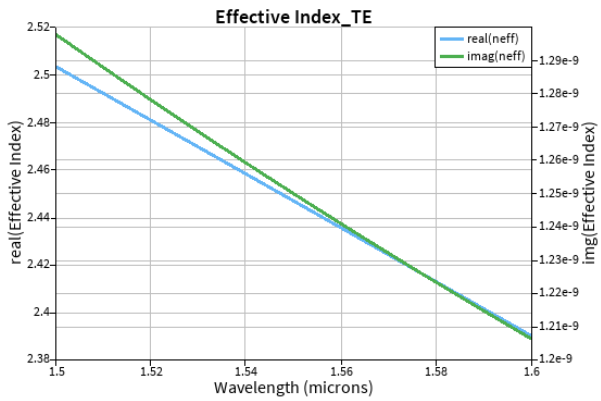

Fig. 4. Relationship of effective index and wavelength for TE mode.

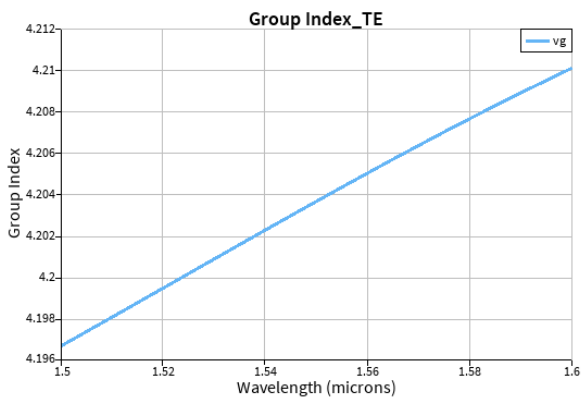

Fig. 5. Relationship of group index and wavelength for TE mode.

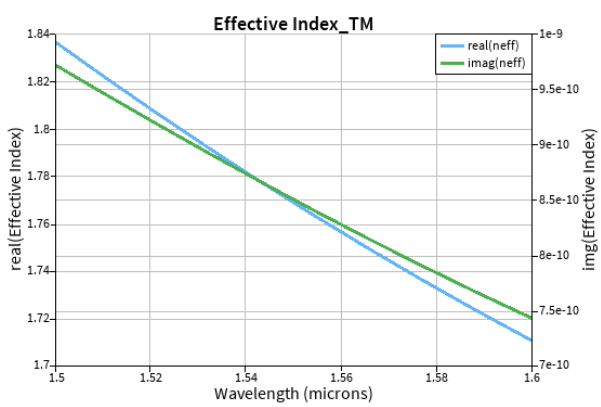

Fig. 6. Relationship of effective index and wavelength for TM mode.

TABLE I
TAYLOR EXPANSION PARAMETERS FOR COMPACT WAVEGUIDE MODEL

	n_1	n_2	n_3
TE	2.44682	-1.13340	-0.04398
TM	1.76882	-1.25870	1.91427

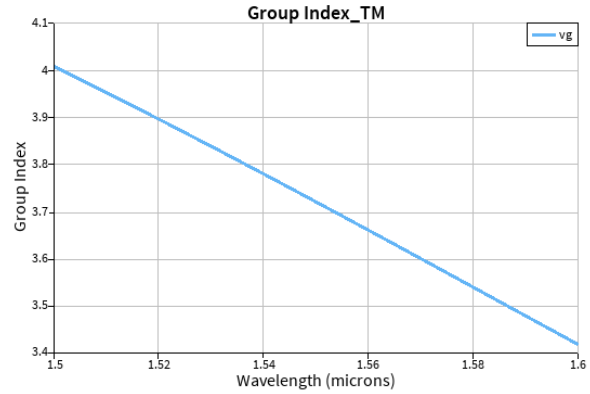

Fig. 7. Relationship of group index and wavelength for TM mode.

TABLE II
LIST OF DESIGN PARAMETERS FOR MACH-ZEHNDER INTERFEROMETER

ΔL [μm]	waveguide width [nm]	FSR_{TE} [nm]	FSR_{TM} [nm]
50	500	11.430	12.915
100	500	5.715	6.458
200	500	2.858	3.229
500	500	1.143	1.292

B. Mach-Zehnder Interferometer

The transfer function of the interferometer versus wavelength can be expressed as:

$$\frac{I_o}{I_i} = \frac{1}{2}\left[1 + \cos\left(\beta_1 L_1 - \beta_2 L_2\right)\right] \tag{2}$$

where I_i is the input light intensity, I_o is the output intensity, and L_1 and L_2 are lengths of the two waveguide arms, and β_1 and β_2 are the waveguide propagation constants, and are defined as:

$$\beta_{1,2} = \frac{2\pi n_{1,2}}{\lambda} \tag{3}$$

For an imbalanced Mach-Zehnder Interferometer with identical waveguide arms but an optical path length difference between these arms (ΔL), Equation 2 becomes:

$$\frac{I_o}{I_i} = \frac{1}{2}\left[1 + \cos\left(\beta \Delta L\right)\right] \tag{4}$$

The free spectral range (FSR) of this kind of imbalanced Mach-Zehnder Interferometer can be expressed as:

$$FSR = \frac{\lambda^2}{\Delta L n_g}[m] \tag{5}$$

Table II lists the design parameters for the imbalanced Mach-Zehnder Interferometer and expected FSR for TE and TM modes.

Figure 8 shows the transmission spectra of a Mach-Zehnder Interferometer with a path length difference of $100\,\mu\text{m}$. The blue curve shows the transmission spectrum without grating couplers, which has approximately zero insertion loss and flat wavelength response. By adding input and output grating couplers in the circuit, the green curve shows a more realistic

simulation where a wavelength dependent insertion loss is present in the transmission spectrum.

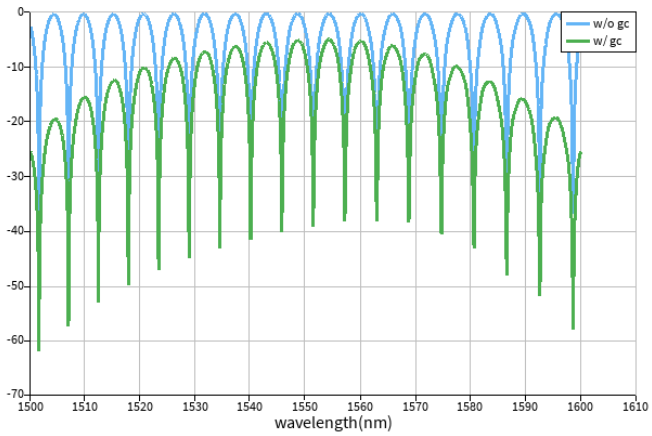

Fig. 8. Transmission spectra [dB] of a Mach-Zehnder Interferometer with ΔL=100 µm with and without grating couplers.